

IoT-enabled circularity tracking and digital product passports for end-of-life PV panel reuse and compliance in European energy companies.

Energy × Regulatory compliance and reporting × IoT platform · OS054 v2 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
54 is the world moving	46 can we play, can we win	HIGH 0 named in the evidence	€171m observed evidence · per year	NEXT regulation adopted or propose...	L1 14 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Evidence gap (SC-13). Orange has 3.0 published reference(s) in Energy, below the threshold of 5. A customer conversation here has no proof point behind it yet.

The opportunity

This opportunity is about deploying IoT-enabled circularity tracking and digital product passports for end-of-life photovoltaic (PV) panels, helping European energy companies comply with upcoming regulations and improve reuse rates. It exists now because the EU is revising grid connection codes and digital product passport mandates are approaching, creating urgency for lifecycle data management.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€207m €46.4m – €2.6bn	€171m €38.2m – €2.2bn	€2.1m €459k – €26.0m	observed
Observed: tendered contract value, annualised	€172m €172m – €460m	€172m €172m – €460m	€2.1m €2.1m – €5.5m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Energy (EU27_2020)	5,796 enterprises	observed	Eurostat · sbs_sc_oww	2024
Enterprises adopting IoT platform	51.7 % of enterprises (10+ employees)	observed	Eurostat · isoc_eb_iotn2 · E_IOT1	2021
Annual value of one engagement	€299k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Size-weighted buyer base	1,341 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

Observed: tendered contract value, annualised

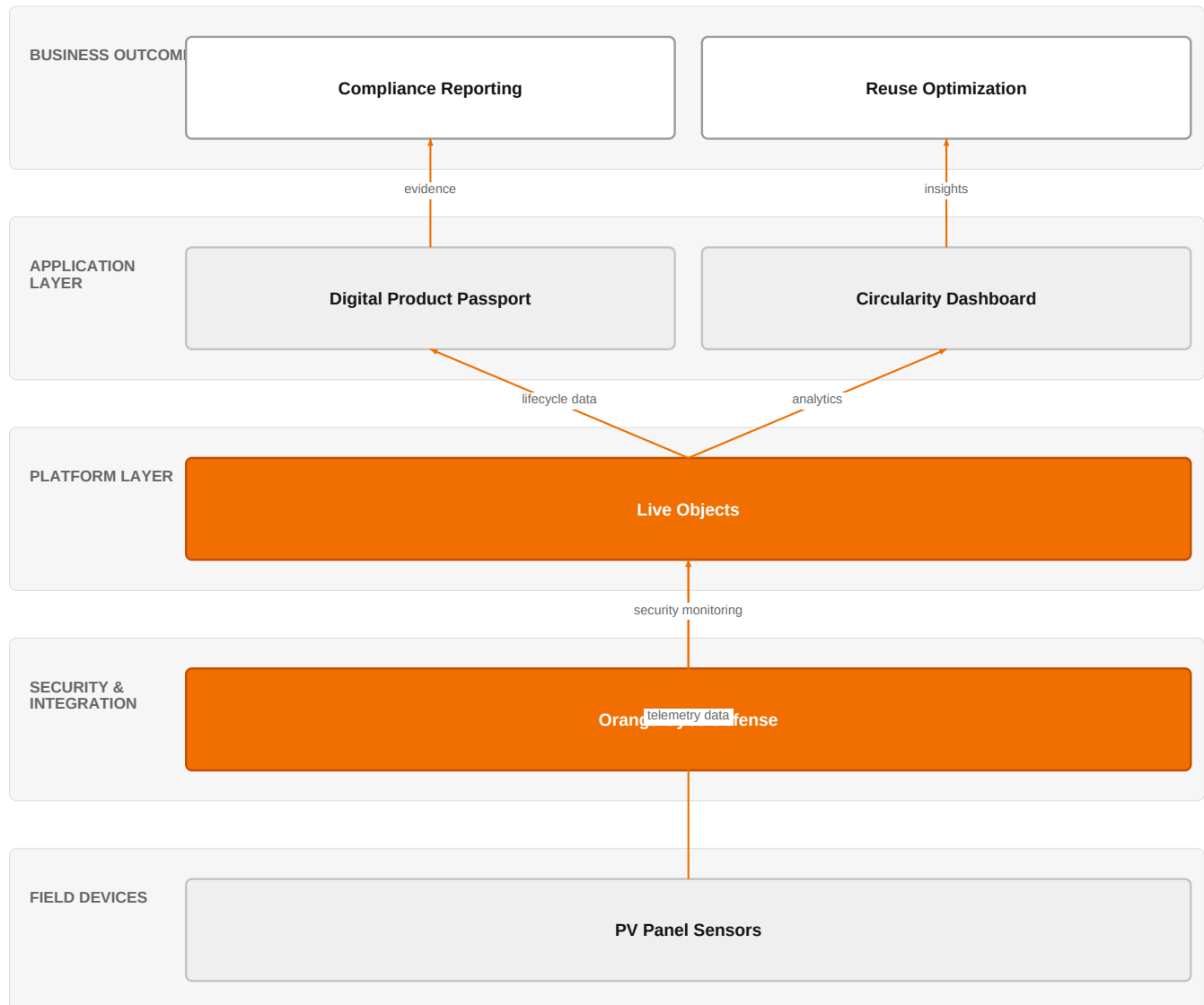
Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€172m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 21 matching notices, matched on vertical via the CPV crosswalk — \$4.5.2's warning applies: a crosswalk error lands directly

in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

The solution, in one picture

PV Circularity Tracking Solution



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This diagram shows how Orange's IoT platform and security services enable end-to-end PV circularity tracking, from sensors to compliance reporting.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

The EU is revising the Network Code on Requirements for Grid Connection of Generators, which will affect energy producers and grid operators, potentially requiring more detailed asset data. Digital Product Passports are mandated by the Ecodesign for Sustainable Products Regulation for electronics and appliances by 2027, creating an urgent need for machine-readable lifecycle data. Projects like SOPHIA are already implementing advanced digital solutions to increase PV panel circularity, indicating market momentum.

Sources: SIG-A8CE3660ACBA ec.europa.eu 2026-07-08; SIG-BA196FF05244 Springer Science and B 2026-02-03; SIG-A2CC51B21A97 cordis.europa.eu 2025-06-01

Who buys, and why now

The buyer is typically the head of sustainability or the chief operating officer at a European energy company with significant PV assets. The pain is operational: they face growing volumes of decommissioned panels with no efficient way to track materials for reuse or prove compliance to regulators. The trigger is an upcoming audit or a new regulation

deadline that forces them to produce lifecycle data they currently lack.

Why it is hot now — every claim bound to a dated source

- The EU is revising the Network Code on Requirements for Grid Connection of Generators, affecting energy producers and grid operators. [SIG-A8CE3660ACBA]
- SOPHIA implements advanced digital solutions to increase circularity of PV panels throughout the full value chain. [SIG-A2CC51B21A97]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution using Live Objects as the IoT platform to collect data from sensors on PV panels, Orange Cyberdefense managed services to secure the data pipeline, and IoT experts to design the tracking architecture. The engagement would start with a discovery phase to map existing assets, then deploy IoT tagging and data collection, and finally integrate with a digital product passport system for compliance reporting.

Why Orange can win

Orange brings a unique combination of IoT expertise, cybersecurity credentials (ISO 27001, SecNumCloud), and a sovereign cloud offering that is critical for handling sensitive energy infrastructure data. The ability to provide end-to-end managed services, from connectivity to security, differentiates Orange from pure IoT platform providers.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Live Objects	offer	L1	offer provides the technology in a matching domain	unconfirmed
Orange Cyberdefense managed services	offer	L1	offer addresses the use case but not this technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
SecNumCloud	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Digital experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

Competitive landscape

HIGH competitive intensity (score 8.2; 0 of 8 listed players appear in this topic's own evidence)

Crowded, with heavyweight incumbents already visible in the evidence. Win on a specific differentiator or do not bid.

Competitor	Category	Position here	Basis	Evidence
AWS	Hyperscaler	sells IoT platform; competes in OX: Smart Industries — also an Orange partner	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	sells IoT platform; competes in Connectivity Solutions, OX: Smart Industries	structural	—
NTT DATA	Global systems integrator	sells IoT platform; competes in Connectivity Solutions, OX: Smart Industries	structural	—
Proximus	Telecom operator peer	sells IoT platform; competes in Connectivity Solutions	structural	—
Telefonica Tech	Telecom operator peer	sells IoT platform; active in Energy; competes in Connectivity Solutions	structural	—
Telia Company	Telecom operator peer	sells IoT platform; competes in Connectivity Solutions, OX: Smart Industries	structural	—

Verizon Business	Telecom operator peer	sells IoT platform; competes in Connectivity Solutions	structural	—
Vodafone Business	Telecom operator peer	sells IoT platform; competes in Connectivity Solutions, OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 How many PV panels do you currently have in operation or decommissioned, and what is your current process for tracking their end-of-life?
- 2 Have you received any regulatory notices or audits that require lifecycle data for your PV assets?
- 3 What is your current data collection method for PV panel materials, and is it manual or automated?
- 4 Do you have a digital product passport initiative underway, or are you waiting for regulation to force it?
- 5 Who in your organization owns the compliance reporting for PV waste, and what is their biggest pain point?
- 6 What is your timeline for needing to demonstrate circularity compliance to regulators or customers?

Objections you will meet

They will say	You can answer
We already have an IoT platform from a hyperscaler; why do we need Orange?	That's true, many platforms exist. But Orange brings the security and sovereignty that energy infrastructure demands, plus managed services that reduce your operational burden.
The regulation isn't final yet; we can wait.	You're right that the details are still evolving, but the direction is clear. Starting now lets you pilot and be ready before the deadline, avoiding a scramble later.
This seems expensive for something we can do with spreadsheets.	Spreadsheets work for small volumes, but as your PV fleet grows and regulations tighten, manual tracking becomes error-prone and costly. Automation pays off in the long run.

Risks and unknowns

The main risk is that regulation timelines slip, reducing urgency. There is also uncertainty about the technical standards for digital product passports and whether customers will accept IoT tagging costs. Additionally, the competitive intensity is high, and Orange's specific PV circularity expertise is unproven.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a European energy producer's PV panel circularity compliance tender, positioning Live Objects as the IoT backbone and ISO 27001 / SecNumCloud certifications as trust anchors, and including Orange Cyberdefense managed services for security.
Sales	In a conversation with the head of sustainability at a European energy producer, say: 'We can help you prepare for the upcoming EU grid connection requirements by using our Live Objects IoT platform to track PV panel circularity and generate the compliance data you'll need.'
Strategist / Innovator	Commission a deep-dive brief on how Live Objects and Orange Cyberdefense managed services could underpin a digital product passport for PV panels, leveraging Orange Group researchers to map the revised EU Network Code requirements to IoT data flows.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-A8CE3660ACBA	2026-07-08	ec.europa.eu	1	Revision of the Network Code on Requirements for Grid Connection of Generators
SIG-B81FDE60EE21	2026-05-01	cordis.europa.eu	1	Smart Upgraded iNverter architectures for Versatile and flexible Integration of PV-Based systEms
SIG-36FFB2435F8A	2026-04-20	openalex.org	1	A Comprehensive Review of Digital Product Passports in Sustainable Manufacturing: Current State...
SIG-36DA075F2A6A	2026-03-02	openalex.org	1	IoT-based Industrial Safety and Monitoring System with Real-Time Alert Mechanism
SIG-A3F10F84C60F	2026-03-01	APL Electronic Devices	1	Industry perspective: Wearable electronic devices for remote patient monitoring
SIG-BA196FF05244	2026-02-03	Springer Science and Busi...	1	Digital Product Passport for Circular Economy: A Review of Semantic Web Vocabularies and Requir...
SIG-CC28A63D0BB7	2026-01-01	openalex.org	1	From Compliance to Circularity: Stakeholder Perspectives on Adopting Digital Product Passports ...
SIG-AECDD651DA98	2025-12-17	openalex.org	1	Edge-AI integrated secure wireless IoT architecture for real time healthcare monitoring and fed...
SIG-955B81B681F1	2025-11-01	cordis.europa.eu	1	INTELLIGENT DIGITAL TOOLS FOR GREEN RENOVATION, ASSESSMENT, TRANSFORMATION, EFFICIENCY AND SUST...
SIG-BCCCB520B0E1	2025-10-21	Institute of Electrical a...	1	Design and Integration of IoT-Enabled Health Sensors for Remote Patient Monitoring
SIG-0D07056A550F	2025-10-01	cordis.europa.eu	1	Persistent Awareness system facilitating Customs Checks
SIG-E6157B37F004	2025-09-14	openalex.org	1	Advancing Circularity in Battery Systems for Renewable Energy: Technologies, Barriers, and Futu...
SIG-75DE7DFC9EA4	2025-06-01	cordis.europa.eu	1	Empowering Circular Operations in the Automotive Plastics Value Chain
SIG-A2CC51B21A97	2025-06-01	cordis.europa.eu	1	Implementation of Advanced Digital Solutions to increase the circularity of PV panels throughou...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding); diagram (1 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:15+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:18+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-18T20:43:36+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.